GEC1044 Chinese Medicine: Theory and Practice



Final Project (Individual)【30%】

Improving the physical and mental health of university students

A Pattern-Based Food Therapy Approach to Nourishment and Everyday Student Wellbeing

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1 Introduction

Periods of limited physical activity taught me how low energy complicates everyday tasks. This motivated a meal that students could prepare with simple equipment and manageable effort. Chinese yam, lotus seed and red date congee addresses the project's concerns—fatigue, poor appetite, disturbed sleep and academic stress—through course-based pattern reasoning adapted to student routines.

The analysis combines this semester's Chinese medicine and dietary therapy knowledge with psychology and modelling from IE2141 Systems Thinking and Dynamics. Figure 1 records pulse-position teaching at Thong Chai. Lecture 7 places pulse findings within a holistic assessment of symptoms, emotions, constitution, routines and environment. Lecture 9 extends this reasoning to dietary selection: the ingredients should match the person's presentation. Together, these principles connect the fieldwork image with the report's design logic—understand the whole presentation, identify the dietary aim, and select ingredients accordingly (GEC1044 2026).

System modelling then connects stress, sleep, fatigue and preparation demands to explore how the recipe fits students' routines (Sterman 2000).

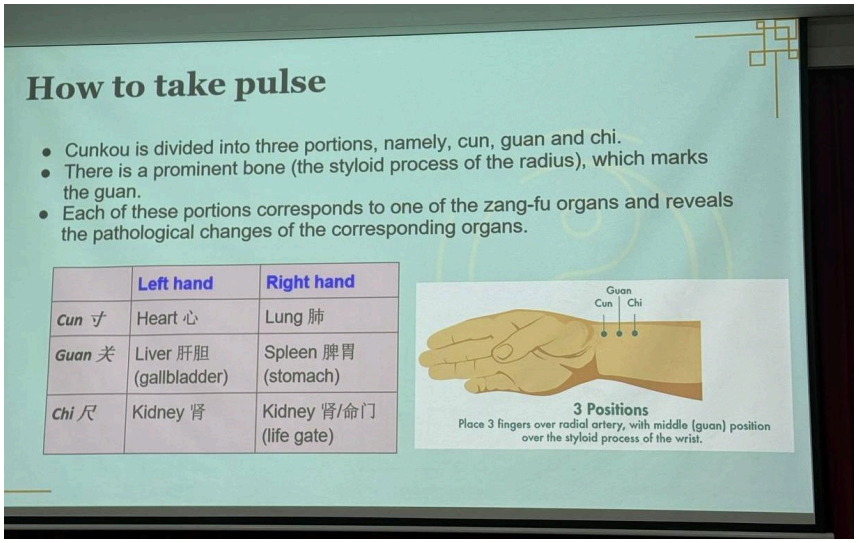


Figure 1. Pulse positions explained during the Thong Chai teaching session. Photograph by the author.

2 Understanding the health concerns

2.1 A Chinese medicine perspective

Lecture 7 distinguishes individual symptoms from Zheng (pattern/syndrome, 证型). Inspection, listening and smelling, inquiry, and palpation bring signs together before a treatment principle is selected.

Fatigue together with poor appetite suggests investigating Spleen Qi deficiency (脾气虚). Within Chinese medicine, the Spleen and Stomach receive and transform food; reduced nourishment gives a traditional explanation for linking eating poorly with low energy. Inquiry would therefore cover meal regularity, stool consistency and whether tiredness persists after rest. The practitioner would also examine the tongue and pulse.

When fatigue and poor appetite accompany interrupted sleep, palpitations and pallor, assessment would consider Heart-Spleen deficiency (心脾两虚). A Chinese University of Hong Kong booklet describes this cluster with loose stools and a pale tongue (官可祈, n.d., 9). Its audience is young cancer survivors; this report borrows the pattern description for its student scenario. The Heart's association with Shen (神), or mental and emotional activity, connects sleep with digestive nourishment.



Figure 2(a). Qi deficiency and dietary matching: the board lists rice, yam and red dates as nourishing food choices.



Figure 2(b). Contrasting guidance for damp-heat and phlegm-dampness illustrates the importance of matching dietary choices to the presentation.

Academic stress connects these digestive and sleep concerns. Lecture 10 distinguishes ordinary emotions from prolonged or excessive emotional strain within the TCM framework, where Heart and Spleen describe functional systems. The HKU School of Chinese Medicine explains that excessive brooding can constrain Qi and impair the Spleen's transformation of food, linking reduced appetite and bloating with diminished nourishment and energy (蔡顯輝 2017). For students worried about deadlines, assessment therefore brings eating, energy and mental rest into the same discussion.

The corresponding food therapy (食疗) principle is to support the Spleen and nourishment while including ingredients traditionally used to calm Shen (神). The Department of Health lists rice, yam and dates for Qi deficiency constitution, a broader category than a current clinical pattern (Hong Kong Department of Health, n.d.-b).

The comparison with other patterns sharpens the recipe's purpose. The University of Hong Kong associates Liver fire insomnia with irritability, and phlegm heat with gastric bloating, a heavy head and bitter taste (梁頌然 2022). These presentations direct attention toward clearing heat or resolving phlegm. Lecture 9 likewise matches dietary properties to the presentation. By contrast, the present recipe follows a nourishing direction for Spleen Qi and Heart-Spleen deficiency. Putting this principle into practice also requires attention to students' routines and emotional strain.

2.2 Psychology and recipe usability

The preceding TCM account connects digestion, rest and emotional strain. A biopsychosocial perspective brings these concerns into the student's daily setting: academic demands and sustained rumination may coincide with disturbed sleep, appetite loss and fatigue, reducing the energy available for study and meal preparation (Hooley, Nock, and Butcher 2021). Figure 3 illustrates the interacting physical, psychological and social influences; Appendix B explains the concepts informing this interpretation.

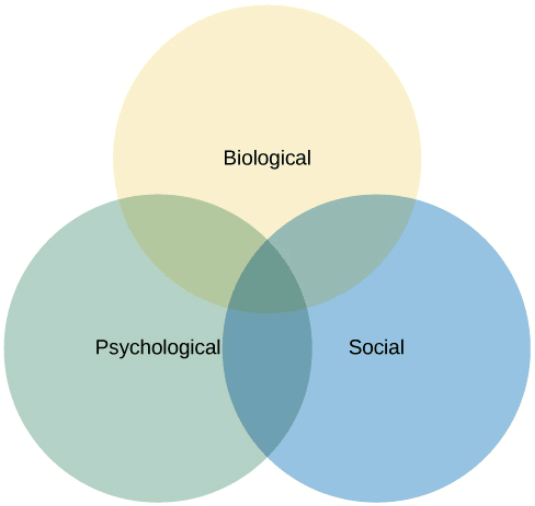


Figure 3. Biopsychosocial model (Spielman, Jenkins, and Lovett 2020, fig. 1.15; CC BY-NC-SA 4.0). Reproduced unchanged; the surrounding discussion applies it to students.

Behavioural activation then provides a practical design principle: make the first task manageable (Cacioppo, Freberg, and Cacioppo 2022). One saucepan, familiar ingredients and two planned portions reduce barriers to starting and repeating preparation. Completing a meal offers a tangible outcome, while scheduling accommodates simmering time. Farmer, Touchton-Leonard and Ross (2018) identify psychosocial benefits in cooking interventions, supporting attention to engagement alongside nutrition. These everyday routines provide the foundation for considering additional individualized care.

2.3 Acupuncture as an additional care option

Beyond food therapy and daily routines, acupuncture can be considered as an additional care option following assessment by a registered TCM practitioner. Lecture 8's distinction between reinforcing deficiency and reducing excess applies the same principle established in Section 2.1: the person's presentation guides the intervention. The Thong Chai teaching photograph (Figure 4) places this individualized approach within a setting that also addresses treatment precautions.

The Thong Chai session notes associate Taichong (LR3, 太冲) with Liver Qi stagnation ("Thong Chai Acupuncture Teaching", n.d.). A study compilation of Ni Haixia's teachings describes bilateral Taichong and Hegu (LI4, 合谷) as the 'Four Gates' ("Acupuncture Quick Reference" 2026, sec. 10). This Qi-regulating example contrasts with the congee's nourishing emphasis, connecting different presentations with different choices. Acupuncture requires a registered practitioner with a valid practising certificate (Ministry of Health 2026). These principles guide the recipe in Section 3.



Figure 4. Acupuncture teaching at Thong Chai. The slide records precautions; the discussion of individualized treatment draws on Lecture 8. Photograph by the author.

3 The recipe and its ingredients

3.1 Chinese yam, lotus seed and red date congee (山药莲子红枣粥)

The starting point is the Hong Kong Department of Health's yam and red date congee: 200 g yam and 30 g red dates with rice (n.d.-a). My version adds lotus seeds, halves the dates for lighter sweetness, and specifies rice and water for repeatable preparation.

The proposed batch makes two portions. All quantities below are planned kitchen weights. Fresh yam and dried ingredients are listed separately because their weights and preparation differ.



Figure 5. Congee prepared in this project: the actual two-portion trial is described in Section 4.5. Photograph by the author.

Table 1. Planned ingredients for two portions

Ingredient	Planned weight	Per portion (half batch)	Specification
Chinese yam 山药	200 g	100 g	Fresh peeled edible portion
Lotus seeds 莲子	20 g	10 g	Dried shelled seeds with green embryos removed
Red dates 红枣	15 g	7.5 g	Dried pitted edible portion
Japonica rice (粳米)	80 g	40 g	Dry, uncooked japonica rice
Water	1,000 ml	500 ml (planned)	Starting amount; add if needed during cooking

3.2 From pattern to food therapy (理—法—方—食)

The design moves from pattern explanation (li, 理), to dietary principle (fa, 法), combination (fang, 方), and preparation (shi, 食). For Spleen Qi and Heart-Spleen deficiency, it aims to strengthen the Spleen and replenish Qi (健脾益气), with nourishment of the Heart and calming Shen (养心安神) for unsettled sleep. Lecture 9's medicine and food sharing a common origin (药食同源) guides this selection. Gui Pi Tang (归脾汤), a classical medicinal formula, similarly combines Spleen support with Heart nourishment (Hong Kong Baptist University, n.d.). This congee adapts that therapeutic direction into everyday food, while medicinal treatment remains individually prescribed.

Design pathway (author's application of Lecture 9)

理: Pattern Spleen Qi / Heart-Spleen deficiency	→	法: Principle Support Spleen; calm Shen	→	方: Combination Yam, lotus seeds, dates, rice	→	食: Preparation Two-portion congee

3.3 Ingredient properties and roles

Chinese yam, shan yao (山药), supplies the digestive foundation. The Hong Kong Polytechnic University (PolyU) describes dried yam rhizome as sweet and neutral, with Spleen, Lung and Kidney affinities and functions supporting the Spleen and Stomach (Hong Kong Polytechnic University, n.d.-d). The fresh-yam target follows the Department of Health's recipe; Figure 6 shows dried reference material.

Lotus seed, lian zi (莲子), links digestive nourishment with mental rest. PolyU lists a sweet and astringent flavour, neutral nature, and Spleen, Kidney and Heart affinities; supporting the Spleen and calming Shen explains its addition to yam (Hong Kong Polytechnic University, n.d.-b). The batch uses 20 g dried, shelled seeds without green embryos.

Red dates, da zao (大枣/红枣), support nourishment and sweetness. Their warm, sweet nature and Spleen–Stomach affinities accompany traditional functions of supplementing middle Qi, nourishing blood and calming Shen (Hong Kong Polytechnic University, n.d.-a). The planned 15 g per batch limits sweetness. Two whole red dates produced a mildly sweet trial without added sugar. This avoids extra carbohydrate while retaining flavour; people with diabetes should count both dates and rice in their portion plan (Centers for Disease Control and Prevention 2024).

Northeast japonica rice (粳米) already available at home provides the congee base. Zhou's dietary teaching classifies japonica rice as neutral and sweet, entering the Spleen and Stomach (周玉弦, n.d., 27). Lecture 9's white/Metal–Lung correspondence is a separate framework. Its traditional nourishing role complements a soft, filling meal; other rice varieties may be used with water and timing adjusted.

Table 2. Traditional ingredient properties and roles

Figure 6. PolyU reference images: dried yam, lotus seeds and red dates; Northeast rice product image supplied by the author.

Ingredient	Nature and flavour	Traditional affinity	Role in this congee
Chinese yam  PolyU reference	Neutral and sweet	Spleen, Lung, Kidney	Appetite and fatigue: main digestive and nourishing focus
Lotus seed  PolyU reference	Neutral, sweet and astringent	Spleen, Kidney, Heart	Appetite and sleep: links digestive support with calming Shen
Red date  PolyU reference	Warm and sweet	Spleen, Stomach	Nourishment and flavour: natural sweetness; 15 g per batch, no added sugar
Northeast japonica rice (东北大米/粳米) 	Neutral and sweet	Spleen, Stomach; white/Metal–Lung correspondence in Lecture 9	Traditionally nourishes Spleen–Stomach; congee base, food carrier, soft texture and satiety

My Thong Chai photograph shows the framework applied here (Figure 7): nature, flavour, meridian affinity and function are separate questions about each ingredient.

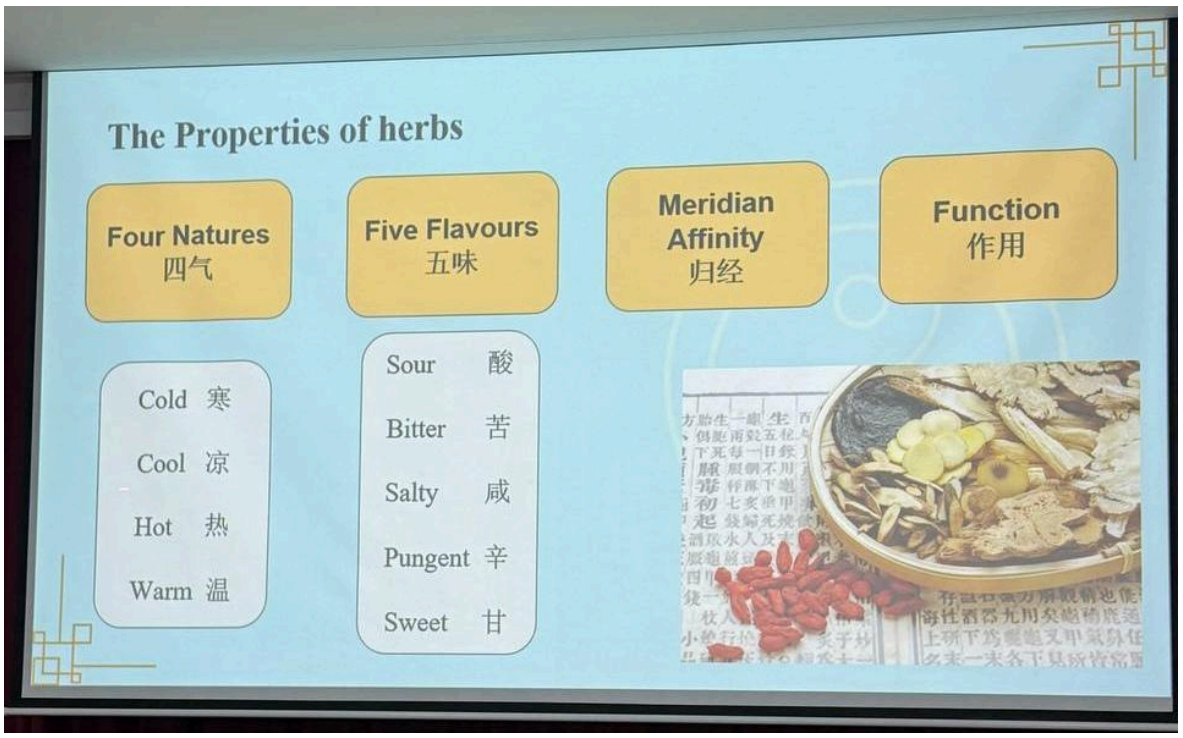


Figure 7. A teaching slide on the properties of herbs at Thong Chai. Photograph by the author.

3.4 Combination and design logic

Using Lecture 9's jun–chen–zuo–shi (君臣佐使) framework, this design assigns yam the chief (jun) role: its Spleen-supporting function addresses the central appetite–fatigue concern. Lotus seed is the deputy (chen), reinforcing that digestive aim while extending the combination toward calming Shen. Dates act as assistant (zuo), adding nourishment and balancing flavour; rice is the carrier (shi), turning the ingredients into a familiar meal. These are the project's design roles, applied to a food combination.

The four ingredients cover the intended digestive and mental-rest functions without expanding the recipe into a medicinal prescription. Additional tonics would need their own pattern and dosage justification. Removing the bitter, cold lotus embryo also keeps the preparation aligned with the nourishing aim and mild flavour (Hong Kong Polytechnic University, n.d.-c). Thus, selection and preparation serve one coherent principle while keeping cooking manageable. In the earlier Tong Keow fieldwork, staff linked lighter qing bu (清补) preferences to Singapore's humid climate (Tutorial Group TD3, Group 2 2026). This recipe likewise favours gentle nourishment, with individual pattern matching guiding selection.

4 Home preparation and practical evaluation

4.1 Cooking reference and preparation plan

The cooking method was adapted from Fu Rui Geng's (福瑞羹) Douyin video, "家里常煮的养胃粥，食材普通又温和" (2026; Figure 8). The narration specifies 60 g rice, 30 g yam, 15 g lotus seeds and two whole red dates.

The video's preparation sequence is:

(1) Soak the lotus seeds in advance until soft.

(2) Put the rice, yam, lotus seeds and red dates into one pot and add enough water.

(3) Simmer over low heat for about 30 minutes until thick and soft. The dates provide sweetness without added sugar.

The trial followed this sequence using Section 3's planned two-portion quantities, with actual adjustments recorded in Section 4.5.



(a) Congee presentation on Douyin



(b) Ingredient preparation on Douyin

Figure 8. Fu Rui Geng (福瑞羹)'s Douyin video: (a) congee presentation; (b) ingredient preparation. Screenshots supplied by the author.

4.2 Ingredient purchase and preliminary cost

The ingredients in Figure 9 were purchased from two local retailers. Giant supplied fresh Chinese yam; Bee's Brand Bird Nest & Health Product Pte Ltd (蜂标燕窝保健品私人有限公司) supplied Xinjiang Ruoqiang pitted red dates (新疆若羌去核红枣) and skinless Hunan Xiang lotus seeds (去皮湖南湘莲子).

The Bee's Brand outlet was at Block 19, Ghim Moh Road #01-251. Package weights and prices are recorded below, with photographs documenting ingredient selection and purchase.

		
(a) Chinese yam (山药) 600 g • S$3.37 Giant	(b) Pitted Ruoqiang dates 若羌去核红枣 500 g • S$4.00 Bee's Brand	(c) Skinless Hunan lotus seeds 去皮湖南湘莲子 250 g • S$5.80 Bee's Brand

Figure 9. Ingredient-purchase photographs: (a) Chinese yam at Giant; (b) pitted red dates; (c) skinless lotus seeds, centre packet. Photographs supplied by the author.

The three packs cost S$13.17; rice was available at home. Allocating the pack prices to the planned quantities gives S$1.7073, approximately S$1.71 per batch or S$0.85 per portion. This preliminary allocation covers yam, lotus seeds and dates. A complete cost record would also include rice, water, energy, peeling losses and measured portion yield.

4.3 Preparation method

For this recipe, use a saucepan with a lid, a kitchen scale and a measuring jug. Weigh the edible portions listed above. Rinse the lotus seeds and follow the packet's soaking instructions, checking that the green embryos have been removed. Wash the rice. Peel the yam with gloves if its sap irritates the skin, then cut it into small pieces. Wash and pit the dates.

Students who prefer to avoid peeling fresh yam can consider packaged dehydrated yam slices, such as Bee's Brand's Dehydrated Rhizoma Dioscoreae (正河南脱水淮山片). Confirm the dry quantity and soaking requirements with the supplier or a registered TCM practitioner, and follow the product's refrigeration instructions. Dried yam requires a separate weight from the recipe's 200 g fresh-yam target (Bee's Brand Bird Nest & Health Product Pte Ltd, n.d.-a).

For the revised method, place the prepared rice, lotus seeds, yam and dates in the saucepan with 1,200 ml water, following the trial's adjustment from the original 1,000 ml plan. Heat until boiling, then reduce to a gentle simmer. Keep the lid slightly ajar, stir occasionally, and cook until the rice opens and the yam and lotus seeds soften. Switch off the heat and rest the congee under a closed lid for five minutes before serving. Section 4.5 records the observed timing and result.

Cool any saved portion quickly, ideally within one hour, refrigerate, and use within 24 hours; reheat once until steaming hot throughout (Food Standards Agency 2026).

4.4 Serving and suitability

Serve one planned portion warm, initially for one or two meals weekly, adjusting to tolerance and routine. Each planned portion supplies 10 g lotus seeds and 7.5 g dates, within the HSA

list's 6–15 g daily medicinal reference ranges (Health Sciences Authority 2025, entries 95 and 230). These inform quantity selection alongside allergies and prescribed diets. The 200 g fresh-yam batch follows a food recipe; dried medicinal weights require separate assessment. Persistent symptoms warrant professional advice.

For people with diabetes or prediabetes, including middle-aged and older readers, replacing some white rice with higher-fibre whole grains such as wholegrain millet (小米), oats or brown rice offers a fibre-rich adaptation. Portion size and cooking method also matter: these grains still contain carbohydrate, and prolonged cooking can increase glycaemic response. Adapt quantities to the person's dietary plan and glucose readings (HealthHub, n.d.-a, n.d.-b).

4.5 Cooking trial and user feedback

The trial used 175 g peeled yam and two whole red dates, with the other ingredients prepared according to the recipe:

(1) Add two complete 600 ml bottles of water (1,200 ml total) and heat on high for about five minutes.

(2) As bubbling approaches the rim, reduce to low heat and leave the lid slightly ajar to reduce boil-over (Figure 10). Simmer for 30–40 minutes until thickened (Figure 10(c)).

(3) Switch off the heat, remove the chopstick used to prop the lid, close the lid and rest for five minutes. Total heating and resting time was about 40–50 minutes.



Figure 10(a). Prepared ingredients and two 600 ml bottles of water. Photograph by the author.

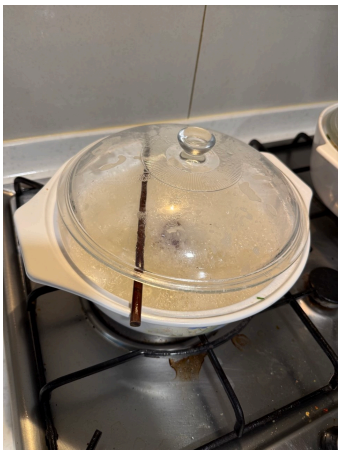


Figure 10(b). Gentle simmering with the lid held slightly open. Photograph by the author.

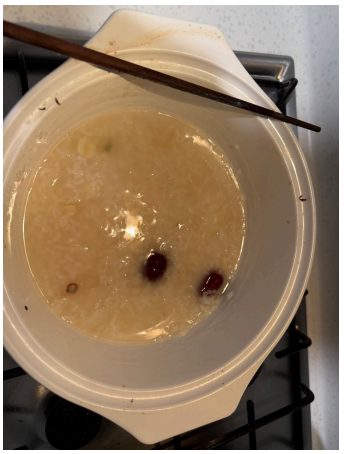


Figure 10(c). Thickened congee before the covered rest. Photograph by the author.



Figure 11(a). Finished congee served in a bowl after the covered rest. Photograph by the author.



Figure 11(b). Another view of the finished congee, showing rice, lotus seeds and a red date. Photograph by the author.

The batch yielded two bowls (Figure 11). The yam and lotus seeds were soft, the consistency was balanced, and the flavour was mildly sweet, partly from the two whole red dates. No extra sugar was added. This mild flavour suits a preference for less-sweet meals. Both students reported comfortable fullness and no post-meal bloating. Together, the observed texture, two-bowl yield and feedback establish the recipe's initial culinary acceptability.

The trial's consistency was acceptable; to test a richer result, the next batch would use 1,000–1,100 ml water, weighed additional red dates and a longer simmer. A one-to-two-week diary could rate preparation effort, appetite, energy and sleep consistently from 1–5, alongside observations of tolerance and routine adherence.

5 Reflection and conclusion

5.1 Modelling approach and results

The 18-week simulation connects academic pressure, rumination, sleep, fatigue, constitution and cooking demands (Lou 2026; Appendix A). In TCM terms, it links Spleen function with Qi–blood generation and nourishment of Shen. Across 1,000 simulated profiles, rounded physical/mental scores rise from 45/48 with unhealthy habits to 67/56 with healthier routines, then 69/59 with weekly congee preparation. Figure 12 includes daily use: further gains accompany greater cooking demands. Lifestyle differences exceed the supplementary congee contribution.

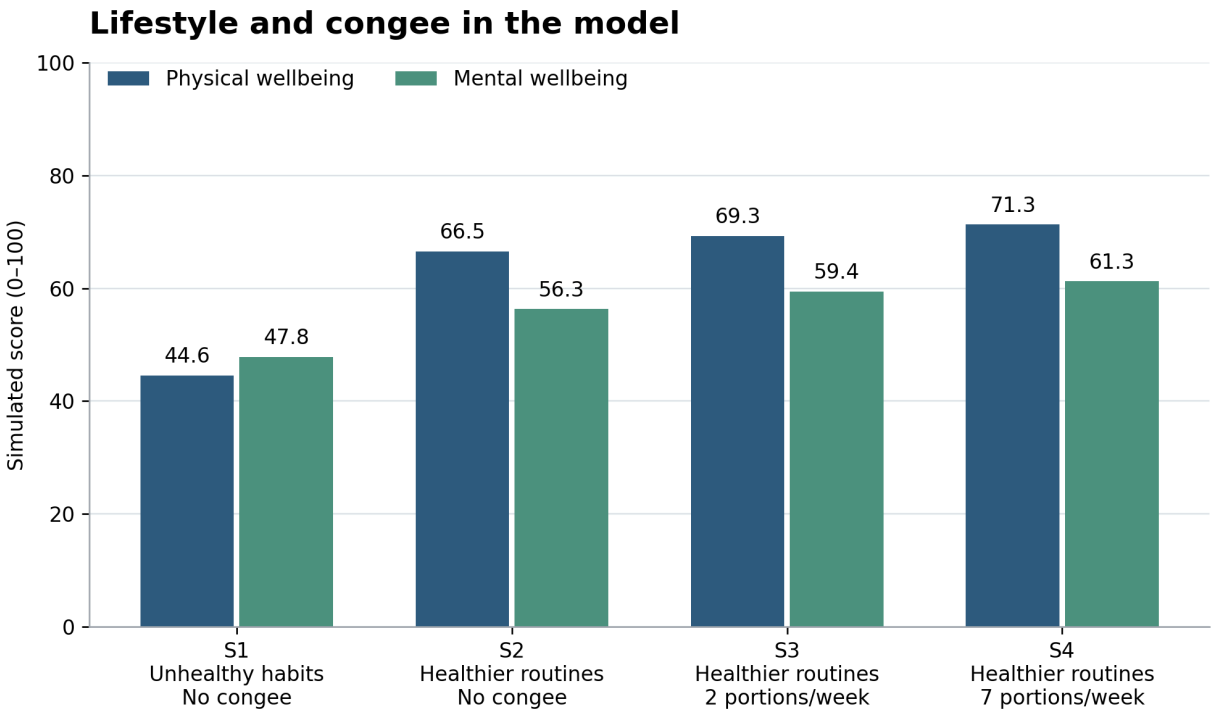


Figure 12. Simulated wellbeing across four scenarios. Medians of profiles' weeks 1–17 average scores; detailed assumptions and outputs appear in Appendix A. Source: Lou (2026).

5.2 Conclusions drawn from the model

The model places food therapy within broader yangsheng (养生): regular meals, rest, emotional balance and suitable activity support the recipe's nourishing aim. Assumed cooking demands rise from about 0.9 hours weekly with weekly preparation to 2.7 hours with daily use. Smaller additional gains accompany this greater burden. The trial supports scheduling two-portion batches; weekly preparation provides two meals, with a second batch an option for further evaluation.

5.3 Overall conclusion

The project connects pattern-guided ingredient selection with an acceptable meal and manageable preparation. Mild sweetness, comfortable fullness and the adjusted cooking endpoint support its practical use within a healthy routine. The evidence has three levels: TCM properties provide the traditional rationale, simulation explores assumed interactions, and two students' feedback establishes initial eating acceptability. Clinical benefit and individual suitability require professional assessment.

Regular hydration, including non-scalding warm water if preferred, supports daily care (NHS 2023). Physically weak students should begin with suitable low-to-moderate activity and avoid high-intensity exertion that aggravates fatigue, seeking medical guidance before increasing intensity (Health Promotion Board, n.d.). Optional acupuncture requires individualized assessment by a registered practitioner with a valid practising certificate (Lecture 8; Ministry of Health 2026).

For severe psychological distress, contact national mindline 1771 first; for physical symptoms, consult a registered TCM practitioner and family doctor. Do not rely on congee alone.

6 Bibliography

“Acupuncture Quick Reference.” 2026. Section 10, “Four Gates: Hegu and Taichong.” Unpublished study compilation of Ni Haixia’s teachings, June 8. Compiler unidentified.

Bee’s Brand Bird Nest & Health Product Pte Ltd. n.d.-a. “Dehydrated Rhizoma Dioscoreae 正河南脱水淮山片 (原切)—400g.” Accessed November 7, 2026. https://www.beesbrand.com.sg/products/dehydrated-rhizoma-dioscoreae-%E6%AD%A3%E6%B2%B3%E5%8D%97%E8%84%B1%E6%B0%B4%E6%B7%AE%E5%B1%B1%E7%89%87-%E5%8E%9F%E5%88%87.

Bee’s Brand Bird Nest & Health Product Pte Ltd. n.d.-b. “Store Locations.” Accessed November 7, 2026. https://www.beesbrand.com.sg/pages/store-locations.

Cacioppo, John T., Laura A. Freberg, and Stephanie J. Cacioppo. 2022. *Discovering Psychology: The Science of Mind*. 4th ed. Cengage Learning.

蔡頴輝 [Cai]. 2017. “思慮太過 損傷脾胃.” University of Hong Kong, School of Chinese Medicine, June 17. https://scm.hku.hk/Views/Media/News/ODN0617.html.

Centers for Disease Control and Prevention. 2024. “Carb Counting.” May 15. https://www.cdc.gov/diabetes/healthy-eating/carb-counting-manage-blood-sugar.html.

Farmer, Nicole, Katherine Touchton-Leonard, and Alyson Ross. 2018. “Psychosocial Benefits of Cooking Interventions: A Systematic Review.” *Health Education & Behavior* 45 (2): 167–180. https://doi.org/10.1177/1090198117736352.

Food Standards Agency. 2026. “Home Food Fact Checker.” Updated August 21. https://www.gov.uk/government/publications/home-food-fact-checker/home-food-fact-checker.

福瑞羹 [Fu Rui Geng]. 2026. “家里常煮的养胃粥，食材普通又温和.” Douyin video, August 20. Accessed November 7, 2026. https://v.douyin.com/9uGQISIwRSk/.

GEC1044. 2026. “Chinese Medicine: Theory and Practice.” Lectures 5 and 7–10. Course teaching materials, National University of Singapore.

官可祈 [Guan], ed., with 林駿炘, deputy ed. n.d. 中醫藥與青少年和青年癌症康復者小冊子：系列一——睡眠困難. Chinese University of Hong Kong. Accessed November 7, 2026. https://scm.cuhk.edu.hk/images/2024/7-6/_01-Booklet__.pdf.

HealthHub. n.d.-a. “Energy-Boosting Wholesome Breakfasts.” Contributed by Health Promotion Board. Accessed November 7, 2026. https://www.healthhub.sg/well-being-and-lifestyle/food-diet-and-nutrition/energy-boosting-breakfasts.

HealthHub. n.d.-b. “Understanding Carbohydrates.” Accessed November 7, 2026. https://www.healthhub.sg/programmes/diabetes-hub/understanding-carbohydrates.

Health Promotion Board. n.d. “Clocking MVPA Minutes.” Section “How to Start Safely.” Accessed November 7, 2026. https://help.hpb.gov.sg/ssp/en/clocking-mvpa-minutes?id=kb_article_view_ssp&sysparm_article=KB0013692.

Health Sciences Authority. 2025. *Reference List of Herbs Used in Single-Ingredient CPM Preparations*. February. Entries 95, 230, and 330. https://file.go.gov.sg/hsa-cpm-single-ing-list.pdf.

Hong Kong Baptist University, School of Chinese Medicine and Library. n.d. “歸脾湯 [Gui Pi Tang].” Chinese Medicine Formulae Images Database, record F00075. Accessed November 7, 2026. https://sys01.lib.hkbu.edu.hk/cmed/cmfid/detail.php?id=F00075.

Hong Kong Department of Health, Chinese Medicine Regulatory Office. n.d.-a. “药补或食补?” Section on Chinese yam and red date congee. Accessed November 7, 2026. https://www.cmro.gov.hk/html/gb/useful_information/public_health/exhibit_boards/Herbal_Tonics_or_Food_Tonics.html.

Hong Kong Department of Health, Chinese Medicine Regulatory Office. n.d.-b. “中醫體質分類與養生.” Boards 2–3, reproduced in Figure 2. Accessed November 7, 2026. https://www.cmro.gov.hk/html/b5/useful_information/public_health/exhibit_boards/Body_Constitution_and_Relevant_Health_Maintenance.html.

Hong Kong Polytechnic University, School of Nursing. n.d.-a. “大枣 [Chinese Date].” Herbal monograph and photograph. Accessed November 7, 2026. https://herbaltcm.sn.polyu.edu.hk/sc/herbal/chinese-date.

Hong Kong Polytechnic University, School of Nursing. n.d.-b. “莲子 [Lotus Seed].” Herbal monograph and photograph. Accessed November 7, 2026. https://herbaltcm.sn.polyu.edu.hk/sc/herbal/lotus-seed.

Hong Kong Polytechnic University, School of Nursing. n.d.-c. “莲子心 [Lotus Plumule].” Herbal monograph. Accessed November 7, 2026. https://herbaltcm.sn.polyu.edu.hk/sc/herbal/lotus-plumule.

Hong Kong Polytechnic University, School of Nursing. n.d.-d. “山药 [Common Yam Rhizome].” Herbal monograph and photograph. Accessed November 7, 2026. https://herbaltcm.sn.polyu.edu.hk/sc/herbal/common-yam-rhizome.

Hooley, Jill M., Matthew K. Nock, and James N. Butcher. 2021. *Abnormal Psychology*. 18th Global ed. Harlow, England: Pearson.

梁頌然 [Liang]. 2022. “中醫藥如何治療失眠.” University of Hong Kong, Faculty of Medicine, February 5. https://www.med.hku.hk/zh-hk/media/knowledge-exchange/newspaper-columns/2022/feb/chinese-medicine-treatment-for-insomnia.

Lou, Yu. 2026. “Model v3 (TCM Holistic): Congee, Cooking Workload, Mindfulness, Liver–Spleen–Heart, Circadian Rhythm, Constitution.” Diagrams, parameter-source table and simulation results. Unpublished GEC1044 project materials, National University of Singapore.

Ministry of Health, Singapore. 2025. “Mental Health Services.” Updated August 21. https://www.moh.gov.sg/seeking-healthcare/find-a-facility-or-service/mental-health-services/.

Ministry of Health, Singapore. 2026. “About.” Sections on TCM registration and service requirements. Updated February 10. https://hpp.moh.gov.sg/tcm-practitioners/about/.

National Institute for Health and Care Excellence. 2022. *Depression in Adults: Treatment and Management*. NG222. “Recommendations.” Accessed November 7, 2026. https://www.nice.org.uk/guidance/ng222/chapter/Recommendations.

National University of Singapore, Office of Student Affairs. 2025. *Student Wellbeing Pulse 2024: Insights on Mental Health, Connection, and Resilience on Campus*. https://osa.nus.edu.sg/wp-content/uploads/2025/03/nus-osa-student-wellbeing-pulse-2024-digital-single-page-updated-1-4.pdf.

NHS. 2023. “Water, Drinks and Hydration.” Last reviewed May 17. https://www.nhs.uk/live-well/eat-well/food-guidelines-and-food-labels/water-drinks-nutrition/.

Spielman, Rose M., William J. Jenkins, and Marilyn D. Lovett. 2020. *Psychology 2e*. Houston: OpenStax. Section 1.3, Figure 1.15, “The Biopsychosocial Model”; reproduced unchanged under CC BY-NC-SA 4.0. https://openstax.org/books/psychology-2e/pages/1-3-contemporary-psychology.

Sterman, John D. 2000. *Business Dynamics: Systems Thinking and Modeling for a Complex World*. Boston: Irwin/McGraw-Hill. https://mitmgmtfaculty.mit.edu/jsterman/business-dynamics/.

“Thong Chai Acupuncture Teaching.” n.d. Unpublished session notes incorporating automated transcription. Compiler unidentified. Project study archive.

Tutorial Group TD3, Group 2. 2026. “Traditional Chinese Medicine at Tong Keow Healthcare.” Unpublished GEC1044 field trip report, National University of Singapore, September 13.

Zheng, Shenglin, Xin Hui Chua, Sarah Martine Edney, Claire Marie Goh, Bee Choo Tai, Janelle Chia, Daphne Koek, Rob M. van Dam, and Falk Müller-Riemenschneider. 2025. “Movement and Dietary Behaviours and Mental Health among University Students: The Health@NUS Study.” *BMC Public Health* 25: 3100. https://doi.org/10.1186/s12889-025-24340-5.

周玉弦 [Zhou]. n.d. 體質與食療. Teaching presentation for the “起動. 靜養. 身心靈” programme, 27–28. Accessed November 7, 2026. https://www.ln-eahmp.hk/upload/files/1/file/6426a2f6e6fde.pdf.

Appendix A: Model Assumptions, Scenarios and Results

A.1 Purpose and evidence

This appendix supplies the assumptions, scenario definitions and numerical outputs behind Section 5 (Lou 2026). The model uses TCM functional relationships to connect nourishment, emotional strain and rest; its numerical settings express the author's interpretation of those relationships. It follows the systems-thinking approach described by Sterman (2000).

A.2 Assumptions and source strength

Table A1. Main assumptions in plain language

Assumption / input	Value used	Source / reason	Evidence strength
Background stress and wellbeing	Targets: 53% elevated stress; mean stress 7.27/16; mean mental wellbeing 54.9/100	NUS Office of Student Affairs 2025; Health@NUS, initially the 2024 preprint (journal version: Zheng et al. 2025)	Survey-informed background; fitted model settings
Lifestyle contrast	Mental-wellbeing gap fitted near 8.5 points; reference association about 8.66 points	Lou 2026, source table; Health@NUS preprint	Observational benchmark; gap partly built into calibration
Time per two-portion batch	75 minutes assumed; varied from 60–90 minutes	Recipe estimate including preparation and clean-up (Lou 2026)	Planning assumption; trial's 40–50 minutes covers heating and resting only
Preparation burden and adherence	Burden setting 0.25, varied ±30%; time-pressure setting 0.60, varied ±30%	Author's judgement about limited time during study (Lou 2026)	Assumed strengths
Stress, rumination and Spleen function	Worry and Liver Qi stagnation each use a 0.10 drain setting, varied ±30%	TCM thinking–digestion relationship (蔡顗輝 2017); Lou 2026	Traditional rationale; numerical effects assumed
Heart nourishment, mental rest and sleep	Influence on worry and sleep: 0.30 each, varied ±30%	Heart–Spleen reasoning (Hong Kong Baptist University, n.d.); Lou 2026	Traditional rationale; numerical effects assumed
Congee contribution	Assigned strength ranges from 0 to 1; diminishing extra effect centred at 3.5 portions weekly	Ingredient rationale and exploratory settings (Lou 2026)	Illustrative assumption; no measured congee effect size
Constitution and variation	Yin-deficiency sleep links ×1.5; Yang-deficiency fatigue build-up ×1.5; many link settings varied ±30%	Constitution scenarios and uncertainty choices (Lou 2026)	Exploratory differences, not diagnosed participant groups

Source: Lou (2026), parameter_sources_v3.csv and model files. Weekly cooking hours also reflect missed servings in the simulation. “Assumed” means the author chose the number; “survey-informed” means published observations guided its selection. A relationship supported by TCM theory still requires an assumed numerical strength. The general physical-wellbeing index is an illustrative model scale; background calibration concerns stress and mental wellbeing.

A.3 Comparison scenarios

Table A2. Four scenarios applied to the same 1,000 simulated profiles

Scenario	Daily habits	Planned congee
S1	Unhealthy routines	None
S2	Healthier routines	None
S3	Same healthier routines as S2	Two portions / one batch weekly
S4	Same healthier routines as S2	Seven portions / 3.5 batches weekly

A.4 Simulation data and diagrams

The model runs for 18 semester weeks. For Table A3 and Figure A1, each profile’s weekly scores are averaged over weeks 1–17; the middle value across the 1,000 profiles is reported. One decimal place preserves the stored output for checking, rather than indicating measurement precision.

Table A3. Semester-average simulated wellbeing

Scenario	Physical score	Mental score
S1	44.6	47.8
S2	66.5	56.3
S3	69.3	59.4
S4	71.3	61.3

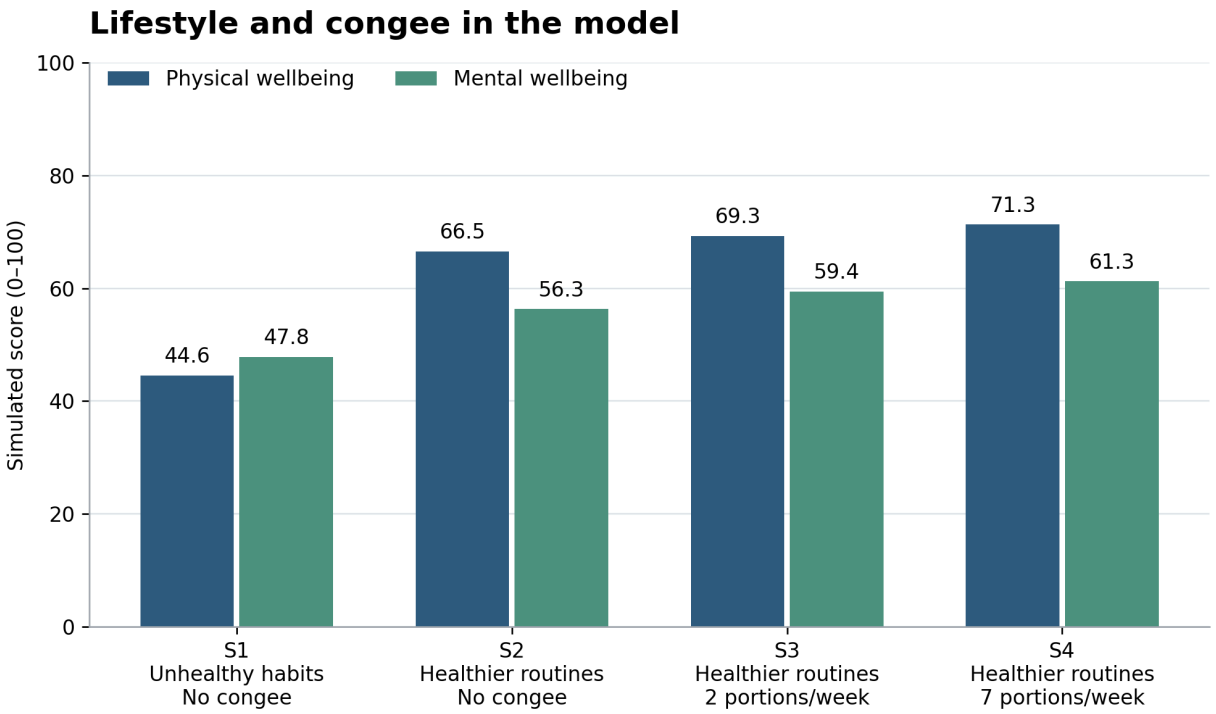


Figure A1. Scenario comparison from the author's model. S1–S4 follow Table A2. Source: Lou (2026), summary_v3.json.

Selected feedback relationships in the model

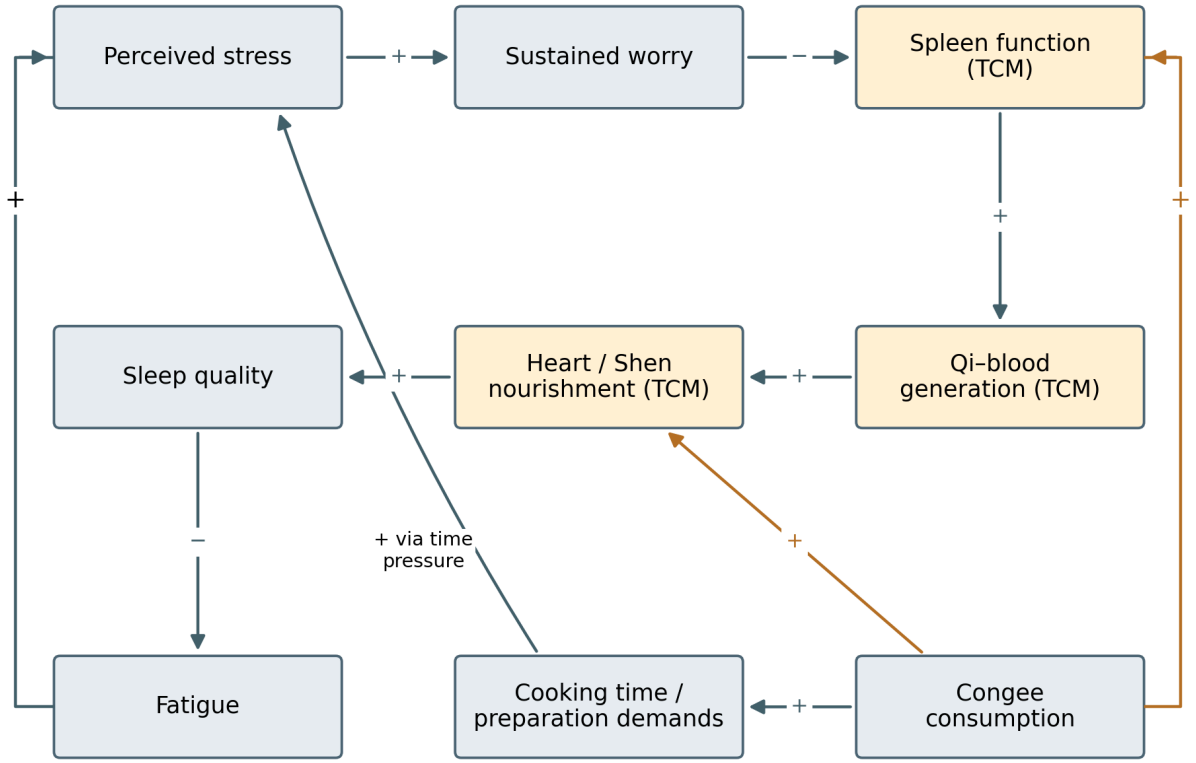


Figure A2. Simplified causal loop diagram, redrawn from the author's model for readability. It shows selected pathways among worry, Spleen function, Qi–blood generation, Shen nourishment, sleep, fatigue and cooking demands. Some arrows combine intermediate steps. Source: Lou (2026).

Selected changing states in the model

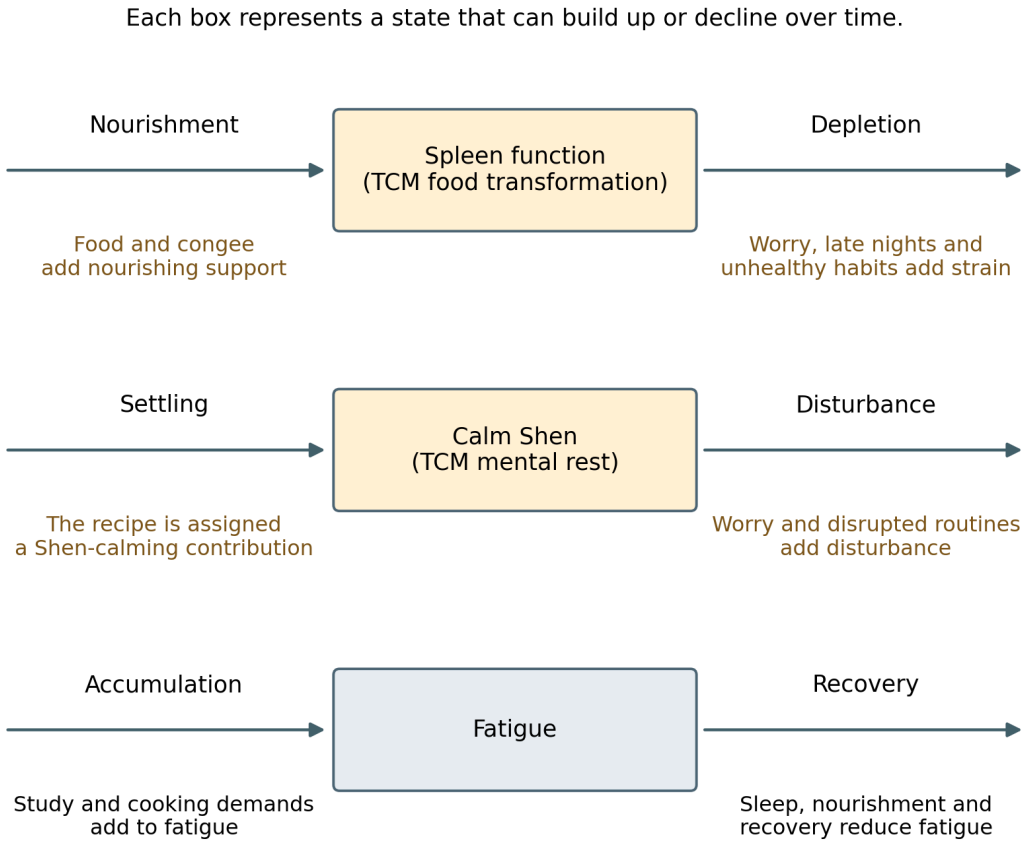


Figure A3. Simplified stock-and-flow extract: nourishment and depletion alter Spleen function; settling and disturbance alter calm Shen; accumulation and recovery alter fatigue. The complete simulation includes additional states. Source: Lou (2026).

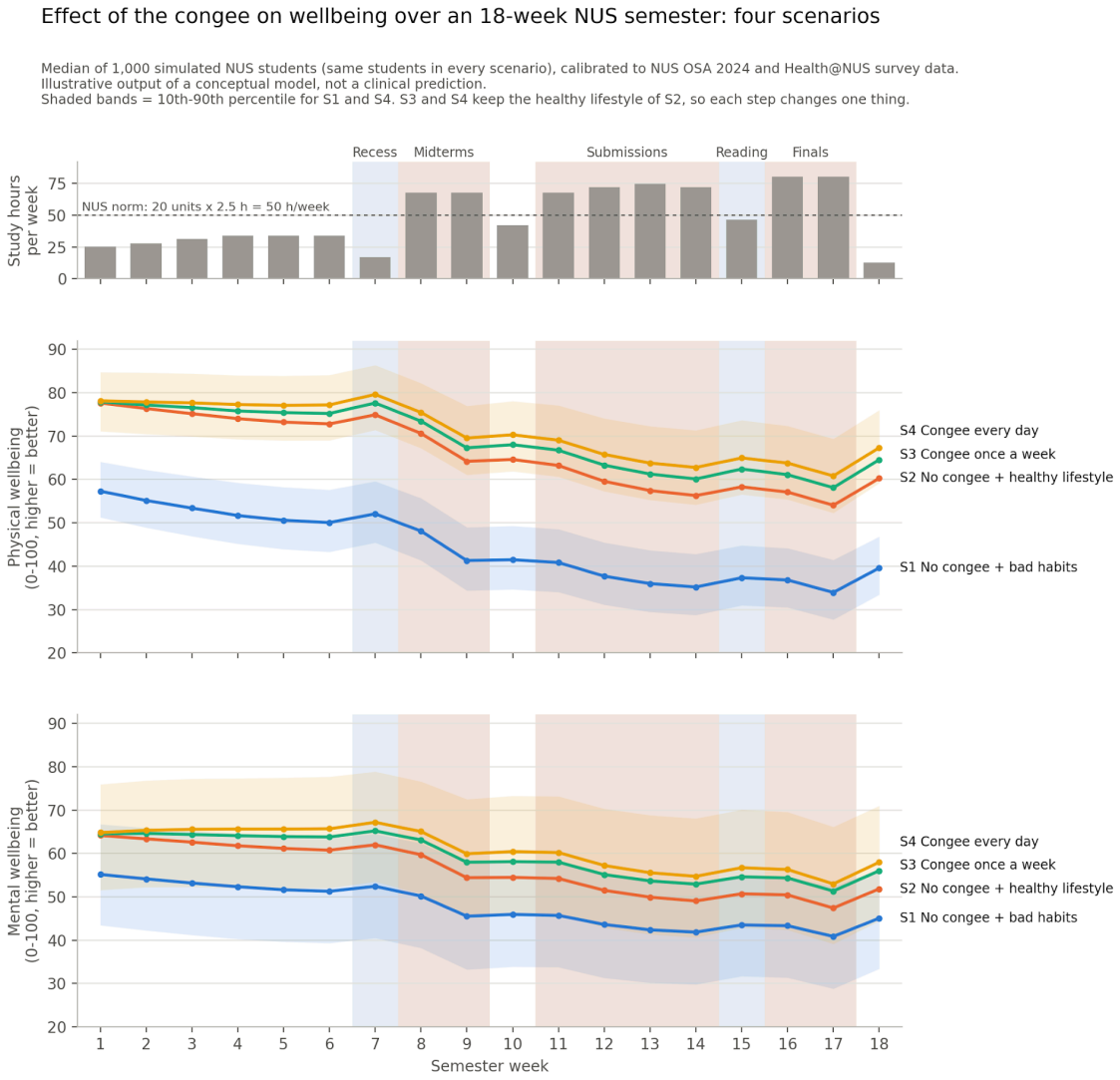


Figure A4. Weekly simulated wellbeing. Lines follow the four scenarios and bands show the 10th–90th percentiles across profiles. Higher study demands coincide with lower modelled wellbeing. Source: Lou (2026), weekly_results_v3.csv.

A.5 Reading the results through TCM

Table A4. Numerical contrasts and traditional interpretation

Model contrast	Numerical difference	Connection with TCM
S2 compared with S1	+22.0 physical; +8.5 mental points	Yangsheng integrates meals, rest, emotions and activity.
S3 compared with S2	+2.8 physical; +3.1 mental points	Focused Spleen nourishment and calming Shen complement daily care.
S4 compared with S3	+2.0 physical; +1.9 mental points; more preparation	Dietary care must fit available energy and time.

Appendix B: Supplementary Biopsychosocial Framework

Table B1. Psychological concepts and project applications

Concept	Definition	Application and boundary
Biopsychosocial framework	Considers interactions among biological, psychological and social influences on health (Spielman, Jenkins, and Lovett 2020, sec. 1.3).	Connects sleep and fatigue, rumination and academic circumstances in a whole-person account of student wellbeing.
Behavioural activation	A structured intervention examining activity–mood relationships and reducing avoidance through constructive activity (National Institute for Health and Care Excellence 2022).	Applies the principle through an achievable cooking task, a tangible outcome and a repeatable routine.
Proposed stress–sleep–fatigue cycle	The project's hypothesis: academic strain and rumination may coincide with disturbed sleep, fatigue and reduced self-care, allowing difficulties to persist (see Figure A2).	Organizes the project's proposed relationship between academic demands, rest, energy and self-care.